

Pleading-to-Proof

Stress-testing a case theory against its own evidence

Team Pleading Compass

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HackTheLaw 2026 · CMS × Harvey Challenge

The litigator's problem

A case is won or lost on one question: **does the evidence actually support what we pleaded?**

- A bundle is hundreds of pages: contracts, emails, witness statements, expert reports.
- Today a lawyer reads it all **by hand** to map every pleaded allegation to a document — slow, late, and easy to miss.
- The worst outcome is an **own-goal**: your *own* bundle contains a document that contradicts your pleaded case — found by the *other side*, at trial.

What a litigator needs, early

Spot weak, unsupported, or self-contradicting propositions **before** trial or settlement — with every finding linked to its source for instant verification.

The challenge: CMS × Harvey “Pleading-to-Proof”

Take a litigation **bundle** + a **pleading** and stress-test the case theory:

- Map each pleaded allegation / defence to the evidence.
- Classify it **SUPPORTED** / **CONTRADICTED** / **unaddressed**.
- Flag contradictions, gaps, and propositions needing more evidence.
- **Keep the lawyer in control** — the tool advises, the lawyer decides.

Prize: a CMS internship. Our front end: **Pleading Compass**.

How a litigator does it by hand (our north star)

From our team lawyer's analysis of the Meridian bundle:

1. Read the pleadings → recover the *case theory*.
2. Build a **chronology** of facts & documents.
3. Evaluate evidence: do the documents *support* or *contradict* the narrative?
4. Weigh witness & expert evidence (and admissibility).
5. Resolve conflicts — *documents prevail over witnesses*.
6. Re-evaluate the pleadings against what the evidence really shows.

Our thesis

This rigorous-but-manual process is exactly what an engine can **systematise** — without taking the judgement away from the lawyer.

Our method: “A2” — a raw-documents → analysis engine

A systematic pipeline (engine_v2):

1. **Split** every document into **numbered paragraphs** — enables clause-level retrieval + verbatim anchoring.
2. **Extract** the pleaded propositions from the pleading.
3. **Retrieve** the top- K paragraphs per proposition (embeddings; instant local fallback).
4. **Detect** contradictions: deterministic surfacers + an LLM classifier.
5. **Verdict**: status + legal overlay + calibrated confidence + a **VERIFY** flag.
6. **Ground** every quote **verbatim** (anti-hallucination).
7. **Emit** an AppData JSON the front end renders.

Two philosophies, blended

(a) LLM-driven judgement

OpenAI for:

- proposition assessment
- contradiction classification
- semantic embeddings / retrieval

Handles nuance, paraphrase, and legal meaning.

(b) Deterministic rigour

Tool-driven guarantees:

- paragraph retrieval
- exact contradiction surfacers
- max-weight graph solver
- Neo4j evidence graph

Exact, reproducible, auditable.

The blend gives both judgement *and* guarantees.

Detecting contradictions: four exact surfacers + an LLM

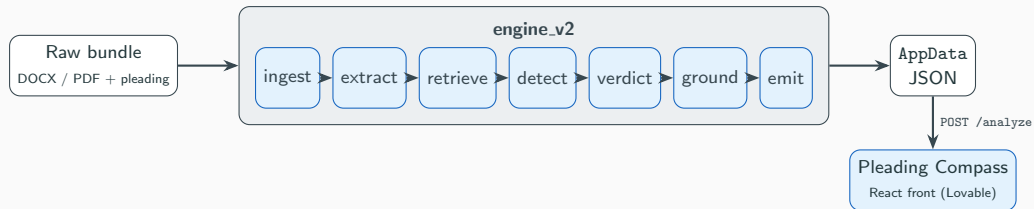
Deterministic surfacers fire *before* the model is trusted:

- **Numeric-disjoint** — “40% of trading hours” vs an expert’s “6.2%”.
- **Supersession** — a later Change Order overrides the date you sue on.
- **Contractual-allocation** — a clause bars or caps the head of loss claimed.
- **Semantic-negation** — “we never accepted” vs a signed acceptance certificate.

The LLM classifier then assesses anything subtler. Result = a verdict with a **legal overlay**:

SUPERSEDED · CONTRACTUALLY_BARRED · CAPPED · CAUSATION_PROBLEM · BURDEN_PROBLEM

Architecture



Front-end views: annotated pleading with **margin verdicts**, an interactive evidence graph, a chronology, and an in-site **source reader** (verbatim highlight over the real PDFs).

Anti-hallucination guarantees

The challenge cares about trust. We give four hard guarantees:

1. **Verbatim quotes only** — every quote is a substring of its cited paragraph, or it is **dropped**. Never invented.
2. **Coverage guarantee** — every pleaded paragraph gets an *explicit* status. Blanks are impossible. (We caught & fixed a silently unannotated paragraph, “02¶6”.)
3. **Calibrated confidence + VERIFY flag** — a blindly-trusting lawyer is, at worst, told “verify this”. Never misled.
4. **Deterministic surfacers** — exact and reproducible, not a model’s opinion.

Own-goals we catch (the Meridian bundle)

Pleaded	Evidence in the same bundle	Verdict
Go-live “fixed at 1 Oct 2024”	Change Order No. 3 moved it to 18 Nov; MSA: time not of the essence	CONTRADICTED
“Meridian never requested a scope change”	Loyalty-module change request & Change Order	CONTRADICTED
“No acceptance was ever given”	Phase 1 UAT <i>Acceptance Certificate</i>	CONTRADICTED
“Unavailable > 40% of trading hours”	Expert report: unavailable $\approx$ 6.2%	CONTRADICTED
Loss of profit “£4.2m”	Expert: $\approx$ £1.3m; MSA bars loss of profit	CAPPED

Each is one click from the [verbatim source](#) that proves it.

Datasets & results

CMS / Meridian (official)

The synthetic bundle, used as our **verified benchmark**.

✓ **13 / 13** propositions correctly resolved.

EU / Brightmarket (ours, GDPR)

A case we built with **7 planted traps**:

- semantic near-miss
- numeric / unit mismatch
- impossible chronology
- broken multi-doc inference chain
- superseded near-duplicate clause
- genuinely-ambiguous claim (⇒ **VERIFY**)
- allocation / burden trap

Automated harness — the hard invariant

A planted-false claim is **never** marked **SUPPORTED** at high confidence — *and* the engine must not “cry wolf” on a clean control.

We meet every ask in the challenge

- ✓ Proposition extraction from the pleading
- ✓ Evidence classification — supportive / adverse / neutral / missing
- ✓ Source-grounded reasoning — verbatim anchors, one-click verify
- ✓ Contradiction & gap detection
- ✓ Human-in-the-loop — **VERIFY** flags; the lawyer decides
- ✓ Case-theory stress test + a **trial-readiness** score

Tools — and a provider-swappable seam

- **OpenAI API** — LLM reasoning + embeddings.
- **Neo4j AuraDB** — the evidence / citation graph.
- **Lovable (React)** — the “Pleading Compass” front end.
- **LaTeX / MiKTeX** — this deck.

One env change, any model

The LLM seam is OpenAI-compatible. **Plug out GPT, plug in a local NVIDIA Nemotron** via a single base-URL / env switch — on-prem, no data leaves the firm.

Honest lineage

- **A1** — teammate Benjamin's backend (same graph idea: claims → evidence graph → LLM robustness). We ran it live on OpenAI; it *inspired* A2.
- **A2** adds the paragraph-split, the contradiction surfacers, verbatim grounding, and the AppData contract.
- A separate seeded **coherence** generator renders a hand-verified *gold* answer key — shown in the demo as the **verified benchmark**.

Transparency

Benchmark = human-verified. Live engine = systematic, and **flags its own uncertainty** on hard cases.

Pleading-to-Proof

From a raw bundle to a source-linked, stress-tested case theory
— and the lawyer stays in control.